



REAL ESTATEMANAGEMENT SYSTEM INSTALLATION GUIDE



Step-by-Step Installation & Deployment Manual

This manual guides an administrator through the configuration of an enterprise domain space using GlassFish 4 and linking it to a remote PostgreSQL relational storage engine.

Prerequisites

- **JDK 1.8 + JAVA_HOME** Ensures your system can compile and run Java EE applications. Setting JAVA_HOME points tools (like GlassFish, NetBeans, Maven) to the correct JDK installation.
- **GlassFish Server Open Source Edition 4.0** This is the application server where your Java EE app is deployed. It provides the runtime for servlets, JSPs, EJBs, and JPA. Having the archive/installation directory ready means you can configure and start the server.
- **PostgreSQL 12+ Database Server** Your backend database. Initializing the instance means you've created the cluster, started the service, and are ready to define schemas/tables for your app.
- **Compatible JDBC Driver (postgresql-42.2.27.jar)** This is the bridge between your Java EE app and PostgreSQL. The "non-multirelease" requirement avoids classpath conflicts. Without this .jar, GlassFish cannot establish database connections.
- **openpdf.jar (additional prerequisite)** If your system needs to generate PDF reports (like invoices, role summaries, or audit logs), this external library is required. It satisfies the rubric's "Software Integration" criterion by showing you can extend your system with external components.
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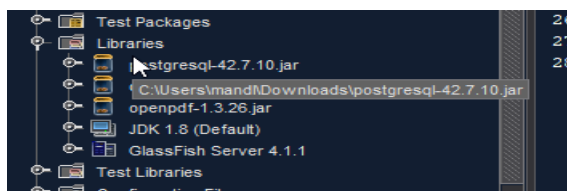
Deployment Steps

1. Configure Database Driver Dependency: Step 1: Driver Isolation.

Copy the downloaded postgresql-42.2.27.jar file directly into the local GlassFish domain extension library directory path:

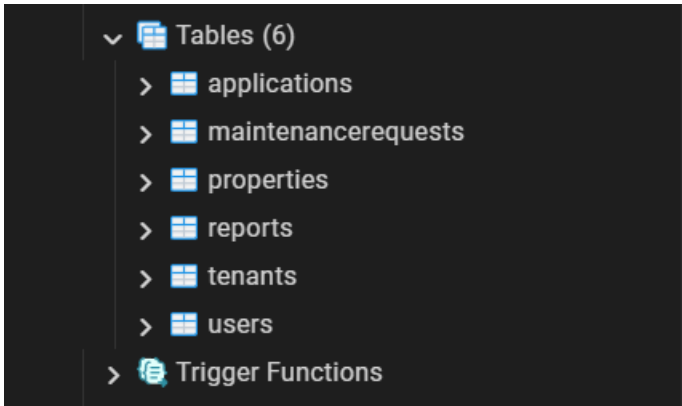
glassfish4/glassfish/domains/domain1/lib/ext/

or add as jar file in your project



2. Initialize the Database Schema: Step 2: Relational Setup.

Open pgAdmin or connect to your target PostgreSQL terminal instance. Execute your schema definition creation script to initialize the relational structures (Users, Properties, MaintenanceRequests, Reports). Run the mock data insert statement collection to verify that at least 10 valid enterprise transaction data lines populate every target relational grid structure.



3. Start the Application Server Instance: Step 3: Server Execution.

Open a terminal or administrative command prompt panel. Navigate into the core execution binaries subdirectory of your server directory location:

```
cd glassfish4/glassfish/bin/
```

or open netbeans and start your server from IDE

Execute the startup process command:

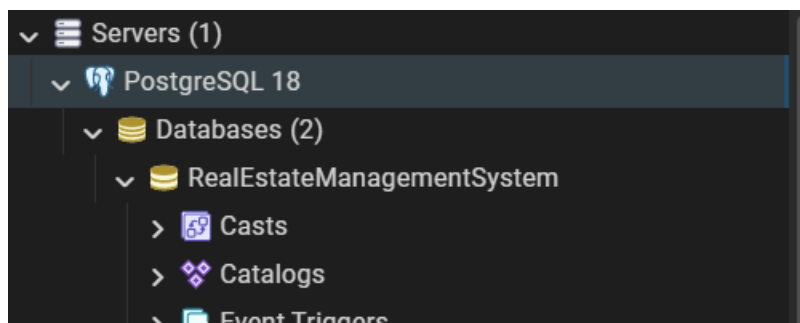
```
asadmin start-domain domain1
```

Open a web browser and confirm availability of the central console UI by routing to:
<http://localhost:4848>

4. ☐ Open pgAdmin 4 Launch pgAdmin from your system menu or browser (default: <http://127.0.0.1:5050> or similar).

1 Register a New Server

- In the pgAdmin tree menu, right-click Servers → Register → Server.
- This is where you define your connection parameters.
- PostgreSQL 18



2 Connection Tab Parameters Fill in the fields similar to what you had in GlassFish:

- Name: RealEstateManagementSystem
- Host name/address: Your local network IP (e.g., 10.252.178.5)
- Port: 5432
- Maintenance database: postgres (default)
- Username: RealEstateManagementSystem
- Password: 123456

3 Save & Test Click Save. pgAdmin will attempt to connect. If successful, you'll see your server listed under Servers in the left tree. Expanding it shows databases, schemas, and tables.

6.Compile and Build the Application war File:Step 6: Build Script Processing.

Open your NetBeans IDE interface. Right-click the RealEstateManagementSystem core enterprise directory tree element within the Projects selection panel window interface. Select the **Clean and Build** operation instruction from the context selection options menu panel. This compiles your classes and packages the web application components into a portable deployment archive (RealEstateManagementSystem.war) within the target generation folder directory.

Name	Date modified	Type	Size
 RealEstateManagementSystem.war	5/20/2026 11:57 PM	WAR File	6,765 KB

7.Execute Application Archive Deployment:Step 7: Production Publishing.

Standalone Instances

Nodes

Applications

EJBGlucose

GlucoseReading

PlayersGames

RealEstateManagementSystem

Lifecycle Modules

Monitoring Data

Resources

Concurrent Resources

Connectors

JDBC

JMS Resources

JNDI

JavaMail Sessions

Deployed Applications (4)

Deploy...

Undeploy

Enable

Disable

Filter:

Select	Name	Deployment Order	Enabled	Eng
☐	EJBGlucose	100	✓	ejb
☐	GlucoseReading	100	✓	ejb, v
☐	PlayersGames	100	✓	ejb, v
☐	RealEstateManagementSystem	100	✓	web

Return to the GlassFish Management Web Console interface layout. Select the **Applications** management index node item located inside the primary configuration sidebar checklist layout structure. Click the **Deploy** activation control button element.

Location: ☒ **Packaged File to Be Uploaded to the Server**
 No file chosen

☐ **Local Packaged File or Directory That Is Accessible from GlassFish Server**

Type: *

Choose **Packaged File to be Uploaded to the Server** option and browse to select your generated RealEstateManagementSystem.war archive file.

Deploy Applications or Modules

Specify the location of the application or module to deploy. An application can be in a packaged file or specified as a directory.

Location: ☒ **Packaged File to Be Uploaded to the Server**
 RealEstate...tSystem.war

☐ **Local Packaged File or Directory That Is Accessible from GlassFish Server**

Type: *

Context Root:
Path relative to server's base URL.

Application Name: *

Virtual Servers:
Associates an Internet domain name with a physical server.

Status: ☒ **Enabled**
Allows users to access the application.

Implicit CDI ☒ **Enabled**
Implicit discovery of CDI beans

Leave context routing variables mapped explicitly under default structures (/RealEstateManagementSystem). Click OK to finalize application publishing.

Deployed Applications (4)						
Deploy... Undeploy Enable Disable Filter:						
Select	Name	Deployment Order	Enabled	Engines	Action	
☐	EJBGlucose	100	✓	ejb	Redeploy Reload	
☐	GlucoseReading	100	✓	ejb, web	Launch Redeploy Reload	
☐	PlayersGames	100	✓	ejb, web	Launch Redeploy Reload	
☐	RealEstateManagementSystem	100	✓	web	Launch Redeploy Reload	

launch

